Muhammad Irfan Fahmi

Computer Engineering | Embedded / IoT Engineering
Klang Valley, Malaysia | cabalme4@gmail.com | irfanfahmi.com/careers/embedded-iot | linkedin.com/in/mifi99 | github.com/l3al3y

Professional Summary

Computer Engineering background at UTeM with practical project work in Arduino-based weighing, load-cell signal handling, hardware/software integration and connected monitoring. Combines C/C++ and Python experience with hands-on technical troubleshooting.

Education

Universiti Teknikal Malaysia Melaka (UTeM), Bachelor of Computer Engineering with Honours — Oct 2022 – Nov 2026

- Graduation scheduled: 2 November 2026.

Politeknik Port Dickson, Diploma in Electronic Engineering (Computer) — Dec 2018 – May 2022

Kolej Komuniti Selandar, Certificate in Computer Systems and Networking — Jul 2017 – Feb 2019

Technical Skills

Embedded & sensing: C/C++, Arduino, HX711 amplifiers, load cells, signal filtering, sensor integration, EasyEDA.

Connected applications: Python, SQL/MySQL, telemetry dashboards, camera/barcode inputs and application integration.

Technical foundations: Hardware diagnostics, software deployment, Windows troubleshooting, VLAN, OSPF and Wireshark; Festo industrial-automation training.

Selected Projects

IoT Livestock Weight Tracking

- Contributed to a team weighing/monitoring platform using Arduino, HX711 load-cell amplifiers, C/C++ signal filtering and telemetry dashboards.
- Team awarded Third Place at INOTEK 2025 (Series 1), 15 January 2025, for Development of Livestock Weight Tracking System Based on IoT.

Hybrid Self-Checkout

- Built a Python pipeline combining YOLOv8/OpenCV product detection, USB barcode input and MySQL product/transaction data.
- Repository-reported evaluation: 77.4% precision and 72.0% recall after 50 epochs; tested visual verification challenges including occlusion and similar packaging.

SMC Crows / Android Release Engineering

- Worked on Android compatibility investigation, release packaging, device testing and a community distribution portal.
- Documented installation and account backup/recovery workflows; release-specific test records explain device and Android-version constraints.

Technical Experience

Technical Staff, Global Elite Ventures Sdn. Bhd — Sep 2021 – Jul 2022

Technical troubleshooting and hardware/software deployment.

Contract Assistant Engineer, ARNN Technologies & Karar Solution — Jun 2020 – Jul 2020

Engineering project support and technical problem-solving.

Technician, OKCS Seri Kembangan — Jun 2018 – Nov 2018

Diagnostics, repairs, hardware maintenance and software configuration.

Professional Registrations

BEM: Graduate Engineer, Computer branch; approved 15 October 2026. Public directory.

MBOT: Graduate Technologist, Information & Communication Technology (IT); GT26100872; approved 22 October 2026.

Selected Training

Festo / STDC: Industrial Automation with AI in Manufacturing, 14-16 July 2026.

Cisco Networking Academy: Enterprise Networking, Security, and Automation; Switching, Routing, and Wireless Essentials (23 February 2026). Course completions.

Languages

Bahasa Malaysia (native); English (professional working proficiency).